

Fig. 2: The Ethernet standard addresses the physical and data link layers of the seven-layer OSI model

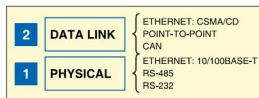


Fig. 3: Industrial Ethernet competes with other communication protocols at the physical and data link layers

Industrial Ethernet competes with several other communication protocols, particularly at physical and data link layers as shown in Fig. 3. Other physical layers include RS-485 and RS-232. Different data link protocols include point-to-point or CAN.

All industrial Ethernet protocols use the same physical layer but add some proprietary features on the data link layer to enable real-time applications. Several other protocol stacks can be added on data link and higher layers, which are responsible for the network, transport and application. Applications such as security management and real-time control can be embedded, resulting in a variety of industrial standards supported by different vendors.

Some of these protocols, such as Ethernet POWERLINK and Modbus, can run on different physical and data layers like 10/100BASE-T and RS-485. Several of them can be supported on

network topologies like bus, tree and star.

Because of the variety of industrial Ethernet flavours, Ethernet is gaining widespread acceptance among equipment manufacturers. In the enterprise environment, Ethernet is primarily implemented in a star configuration. However, some industrial Ethernet protocols allow bus and tree topologies.

From a security and reliability standpoint, star configurations provide some advantages since these do not affect other devices on the star in case of failure. With bus and tree topologies, this is not always the case.

In the industrial world, bus topologies have been historically quite popular. Therefore some protocols like EtherCAT and PROFINET enable bus configurations on top of the Ethernet protocol.

PoE could power industrial applications soon

PoE is becoming widely available on enterprise networks where it remotely powers the edge of the network close to applications. Today, it powers VoIP phones and WLAN access points and is becoming popular in industrial security applications such as security cameras, access control systems and RFID access points.

For industrial networks, the network edge in every node at the end of the industrial network is close to the manufacturing line or process. In parallel with adoption in the enterprise world, industrial applications are becoming PoE-enabled. Some industrial Ethernet switches are available with PoE-enabled ports. A few WLAN access points for industrial networks and

small industrial cameras powered over the industrial Ethernet network are available in the market.

This is only the first step; there are more opportunities for power over industrial Ethernet. Other possible elements to supply with PoE networks are gateways, sensors, interfacing devices such as HMI, and, potentially, parts of PLCs, drives and I/O modules.

PoE depends on two devices, situated at the opposite ends of the cable. The power sourcing equipment (PSE) is normally an industrial Ethernet switch or zone controller and distributes the power on the cable. On the consumer side, the powered device (PD)—such as WLAN access point, sensor, I/O or PLC—accepts the power and converts it into the voltage level required by the application (refer Fig. 4).

Is PoE ready for industrial adoption?

Several criteria drive adoption of new technologies in the industrial world. Flexibility, safety, integration, support, price, compliance and robustness are the dominant ones. By itself, industrial Ethernet aligns well with these criteria, and adding PoE to a network only helps to further satisfy these requirements.

Industrial equipment must be easily programmable and flexible for changes in configuration. Bringing power over industrial Ethernet can remove power plugs and converters, reduce cabling cost, and simplify reconfiguration or new equipment additions.

Robustness and reliability are key requirements for industrial equipment. PoE in combination with an uninterruptible power supply on the Ethernet switch can significantly improve plant safety in the event of a power failure. When power fails, all equipment on the Ethernet cable stay active. This provides the ability to monitor critical functions and trigger alarms, as well as safely manage the power restoration process.

Reliability of the PoE network may be a concern, particularly in harsh